ERICK OLUGA ABOGE

Full-Cycle Software Engineer

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GitHub • LinkedIn • Portfolio

PROFESSIONAL SUMMARY

Full-cycle software engineer with 3+ years of experience building production systems from architecture to deployment. Proven ability to own projects end-to-end while collaborating effectively in team environments. Delivered enterprise systems including payment integrations handling real transactions (90% error reduction), workforce compliance platforms, and multi-tenant ESG reporting systems on AWS. Strong foundation in backend development, infrastructure, testing practices, and reliability engineering.

TECHNICAL SKILLS

Backend & APIs: Python, Django, Node.js, Express, C# .NET Core, REST APIs, PostgreSQL, MySQL, Redis

Frontend: React, TypeScript, Next.js, TailwindCSS, Zustand, Framer Motion

Infrastructure: Docker, AWS (App Runner, Amplify, RDS, S3, SES), GitHub Actions, CI/CD, Linux

AI/ML: LangChain, LLM Integration, AI Agents, Transformers.js

Architecture: Clean Architecture, Domain-Driven Design, RBAC, Multi-tenancy, Payment Integration

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer — Optiven Limited | Nairobi, Kenya | December 2024 – Present

- Architected enterprise CRM managing 10,000+ customer records with server-side pagination and database indexing, reducing query time by 40%
- Built frontend for HRMS Performance Module in collaboration with backend team — 8-state approval workflow serving 150+ employees
- Integrated Microsoft Dynamics 365 via C# .NET APIs for real-time financial reporting on executive dashboards
- Containerized applications with Docker ensuring consistent environments across development, CI, and production
- Collaborated with 5-person development team using Git workflows, code reviews, and agile practices

Contract Software Developer — Optiven Limited | Nairobi, Kenya | August 2024 – December 2024

- Designed and deployed Daily Work Register system end-to-end (React/TypeScript, Node.js), replacing manual Excel processes for entire workforce
- Led M-Pesa payment integration handling real transactions — designed for failure with explicit timeouts, idempotent webhooks, and transaction state tracking. Reduced reconciliation errors by 90%
- Implemented Role-Based Access Control (RBAC) with automated email notifications for Employee, Supervisor, and HR dashboards
- Converted from contract to full-time based on delivery quality and system reliability

Software Developer (Contract) — GMC Kitengela | Nairobi, Kenya | January 2024 – August 2024

- Engineered end-to-end Procurement Module for Point of Sale system using React and Django
- Digitized supply chain workflow: Requisition → RFQ → Bid Analysis → Purchase Order → GRN → Invoice
- Reduced procurement approval turnaround time and improved inventory accuracy across operations

Junior Software Developer — Freelance (Upwork) | Remote | November 2022 – January 2024

- Completed 15+ projects with 5-star rating — e-commerce websites, finance applications, news platforms
- Built frontend applications for international clients using React and modern JavaScript frameworks
- Developed backend services with .NET for payment gateway integrations

KEY PROJECTS

Enterprise ESG Reporting Platform | React, TypeScript, Node.js, PostgreSQL, AWS

- Full-cycle ownership of enterprise ESG data collection system — 150+ KPIs, two-level approval workflow, PDF reporting
- Deployed on AWS infrastructure: Amplify (frontend), App Runner (backend), RDS, S3, SES for emails
- Live Demo: main.dwt9oplenis7k.amplifyapp.com

MiniDB — Relational Database Engine | Python, pytest

- Built database from scratch: SQL lexer, recursive descent parser, query executor, B-tree indexing
- 49 automated tests ensuring correctness. Demonstrates CS fundamentals beyond framework usage
- Live Demo: n5t32y4qnz.eu-west-1.awsapprunner.com | GitHub: github.com/Abogeerick/minidb-complete-project

Nexus Financial — AI-Powered Finance Platform | Next.js, TypeScript, Prisma, Framer Motion

- Personal finance platform with M-Pesa parsing, AI transaction categorization, and interactive analytics
- Live Demo: nexus-app-plum.vercel.app | GitHub: github.com/Abogeerick/nexus-app

AI-Powered Research Agent | Python, LangChain, Streamlit

- LLM-powered agent that searches the web and synthesizes answers with citations using RAG pipeline
- GitHub: github.com/Abogeerick/Ask-the-web

EDUCATION

Bachelor of Commerce — Finance | Second Class Upper Division
University of Nairobi | September 2020 – December 2024

Certificate in Software Development
Moringa School | May 2022 – November 2022

ADDITIONAL

Languages: English (Fluent), Swahili (Native)